

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

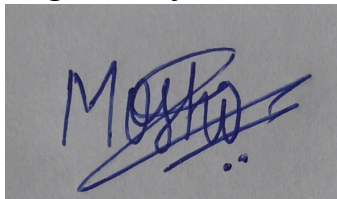
SCENARIO-BASED QUESTIONS

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

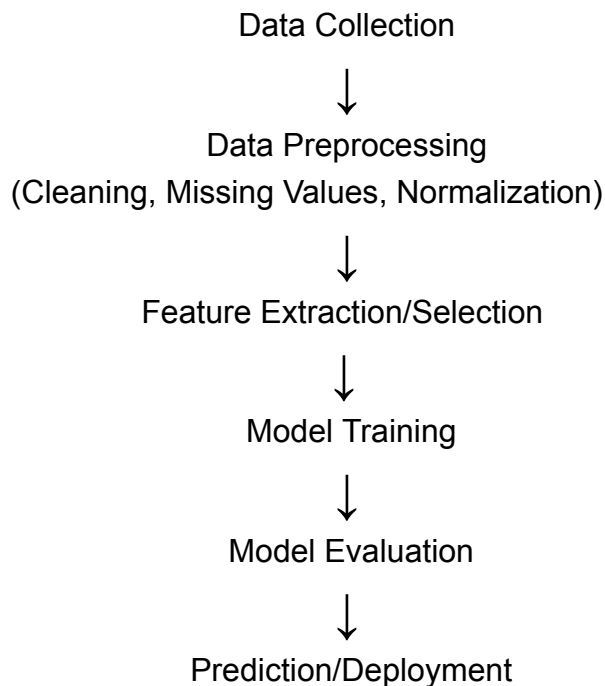
A handwritten signature in blue ink, appearing to read 'Moshika', is written over a grey rectangular background.

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

ANSWER:



Explanation

1. Data Collection

- Data is collected from sensors, databases, websites, IoT devices, etc.
- Example: Student marks, medical records.

2. Data Preprocessing

- Removes missing values and duplicates.
- Normalizes data.
- Converts categorical data into numerical values.

3. Feature Extraction

- Selects important attributes.
- Example: From customer data, choose Age, Income, Purchase History.

4. Model Training

- Machine learning algorithm learns patterns.
- Example: Decision Tree, Neural Network, SVM.

5. Model Evaluation

- Performance measured using Accuracy, Precision, Recall and F1-score.

Supervised Learning

- Uses **labelled data**.
- Example:

Image	Label
Cat Image	Cat
Dog Image	Dog

The model learns from known outputs.

Unsupervised Learning

- Uses **unlabelled data**.
- Example:

Customers grouped into similar purchasing patterns using **K-Means Clustering**.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

ANSWER:

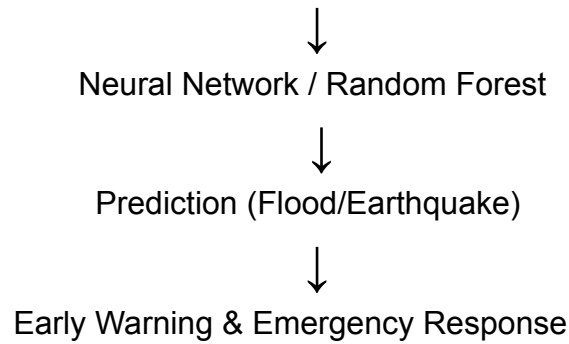
Historical Data + Real-Time Sensor Data



Data Preprocessing



Feature Selection



Data Sources

- Historical rainfall records
- River water levels
- Weather forecasts
- Earthquake sensor data
- Satellite images

Suitable Model

Artificial Neural Network (ANN)

Reason

- Handles nonlinear relationships.
- Learns complex weather patterns.
- Gives accurate predictions.

(Random Forest can also be used because it handles mixed data and reduces overfitting.)

Preprocessing

- Remove missing values
- Normalize data
- Remove duplicates
- Encode categorical values

Feature Selection

Important features:

- Rainfall
- Temperature
- River level
- Soil moisture
- Seismic activity

Training

- Train using historical data.
- Validate using validation dataset.
- Test on unseen data.

Evaluation

Metrics:

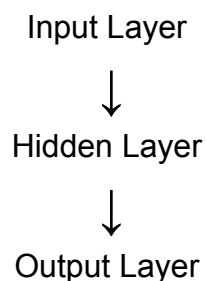
- Accuracy
- Precision
- Recall
- F1-score

Applications

- Early flood warning
- Earthquake risk prediction
- Rescue planning
- Resource allocation
- Emergency response

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

ANSWER:



Steps

1. Forward Pass

Inputs pass through hidden layer to output.

Example:

Input = 0.8

Predicted Output = 0.70

Actual Output = 1

2. Error Calculation

Error = Actual – Predicted

= 1 – 0.70

= 0.30

3. Backpropagation

Error travels backward.

Weights updated using

$$\text{New Weight} = \text{Old Weight} - \eta \times \text{Gradient}$$

where η = Learning Rate

Learning Rate

- High $\rightarrow$ Fast learning but unstable.
- Low $\rightarrow$ Slow but stable.

Activation Functions

- Sigmoid
- ReLU
- Tanh

They introduce non-linearity and improve learning.

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

ANSWER:

Initial Population

The algorithm starts by generating an initial set of possible solutions (population). Each solution is represented by different characteristics (genes) such as speed and accuracy, along with a fitness score that indicates its quality.

Example	Speed	Accuracy	Fitness
1	High	Low	40
2	Medium	Medium	60
3	High	Medium	75
4	Low	High	65
5	Medium	High	80

Selection

The selection process identifies the fittest individuals to become parents for the next generation. Solutions with higher fitness have a greater chance of being selected.

Selected Parents:

- Example 5 → Fitness = **80**
- Example 3 → Fitness = **75**
- Example 4 → Fitness = **65**

These individuals are chosen because they represent better-quality solutions.

Crossover

Crossover combines the characteristics of two parent solutions to produce new offspring. The goal is to inherit the best traits from both parents.

Parents:

- **P1:** Medium Speed, High Accuracy
- **P2:** High Speed, Medium Accuracy

Offspring:

- **Child 1:** High Speed, High Accuracy
- **Child 2:** Medium Speed, Medium Accuracy

The first child combines the strongest traits from both parents, resulting in a higher fitness value.

Mutation

Mutation randomly changes one characteristic (gene) of an offspring. This helps maintain diversity in the population and prevents the algorithm from getting stuck in a local optimum.

Example Mutation:

- Speed: **Medium** → **High**

A small random change can create a better solution that was not present in the previous generation.

Final Best Solution

Speed	Accuracy	Fitness
High	High	90 (Optimal)

After several generations of selection, crossover, and mutation, the algorithm converges to the best solution with the highest fitness score.

Genetic Algorithm Operations

- **Selection:** Chooses the fittest individuals for reproduction based on fitness values.
- **Crossover:** Combines parent traits to generate improved offspring.
- **Mutation:** Introduces random changes to maintain diversity and discover better solutions.
- **Result:** Repeating these operations over multiple generations gradually improves the overall quality of the population and leads to an optimal or near-optimal solution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs.

(5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

(5 Marks)

ANSWER:

Dataset Analysis

Example	Actual	Predicted	Result
1	Yes	Yes	TP
2	Yes	No	FN
3	No	No	TN
4	Yes	Yes	TP
5	No	Yes	FP

Confusion Matrix

TP = 2

TN = 1

FP = 1

FN = 1

Accuracy

Accuracy

= (TP+TN)/(Total)

$$= (2+1)/5$$

$$= 3/5$$

$$= 60\%$$

Precision

$$TP/(TP+FP)$$

$$=2/(2+1)$$

$$=66.67\%$$

Recall

$$TP/(TP+FN)$$

$$=2/(2+1)$$

$$=66.67\%$$

F1-Score

$$2PR/(P+R)$$

$$=66.67\%$$

Preprocessing

- Missing value removal
- Normalization
- Encoding categorical values

Feature Selection

Important features:

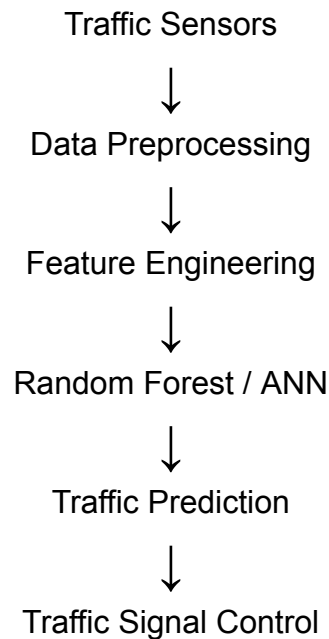
- Age
- Blood Pressure
- Glucose
- Lifestyle Risk

Benefits:

- Improves accuracy
- Reduces overfitting
- Faster training

6. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

ANSWER:



Input Features

- Vehicle density
- Time
- Weather
- Road type

Suitable Algorithm

Random Forest

Reason:

- High accuracy
- Handles mixed data
- Resistant to overfitting

Preprocessing

- Missing value handling
- Scaling
- Encoding

Feature Engineering

- Peak hour
- Rain indicator

- Holiday flag

Training

- Split into Train, Validation and Test datasets.

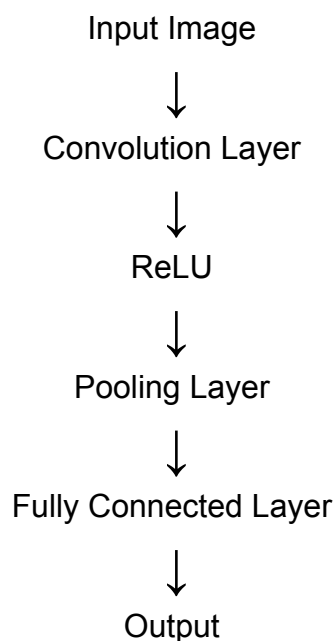
Deployment

- Live traffic prediction
- Smart signal timing
- Route optimization

7. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

ANSWER:

CNN Architecture



Layers

Convolution Layer

- Detects edges, shapes and textures.

Pooling Layer

- Reduces image size.
- Max Pooling is common.

Fully Connected Layer

- Performs final classification.

Feature Maps

Generated by applying filters over the image.

Backpropagation

- Calculates error.
- Updates filter weights.

Example

Image Classification:

Input Image → CNN → Cat/Dog

Factors Affecting Performance

- Dataset quality
- Number of layers
- Learning rate
- Batch size
- Number of epochs

8. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

ANSWER:

Chromosome Representation

Age BP Glucose BMI

1 1 0 1

1 = Selected

0 = Not Selected

Fitness Function

Measures:

- Accuracy
- Number of selected features

GA Process

Selection

- Choose highest accuracy subsets.

Crossover

- Combine two feature subsets.

Mutation

- Flip one bit.

Example:

1011



1111

Advantages

- Removes irrelevant features
- Reduces computation
- Improves model accuracy
- Avoids overfitting

9. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

ANSWER:

Sample Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	50	10
Actual No	5	35

TP=50

TN=35

FP=5

FN=10

Accuracy

$$(50+35)/100$$

$$=85\%$$

Precision

$$50/(50+5)$$

$$=90.9\%$$

Recall

$$50/(50+10)$$

$$=83.3\%$$

F1-score

$$2 \times 90.9 \times 83.3$$

$$90.9+83.3$$

$$\approx 86.9\%$$

ROC Curve

Shows relationship between:

- True Positive Rate
- False Positive Rate

Applications

- Medical diagnosis
- Fraud detection
- Spam filtering
- Credit approval

10. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosome and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently.

(5 Marks)

ANSWER:

Chromosome Encoding

Task Order

A B C D E

Each chromosome represents one scheduling solution.

Fitness Function

Based on:

- Minimum completion time
- Resource utilization
- Minimum waiting time

GA Steps

Selection

- Select schedules with higher fitness.

Crossover

Example:

Parent1

A B C D

Parent2

D C B A



Child

A B B A

(then repair duplicates if needed)

Mutation

Swap two tasks.

A B C D



A C B D

Advantages

- Finds near-optimal schedules quickly
- Avoids exhaustive search
- Handles large scheduling problems efficiently
- Produces high-quality task allocations

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

